CSE276 — AI FOUNDATIONS I WEEK 6 I LECTURE 12

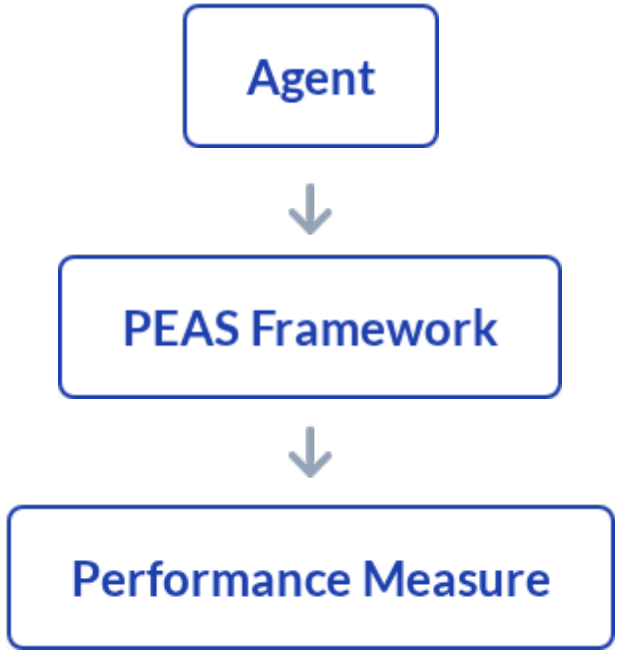
Rational Agents & Environment Properties

How should an intelligent agent behave — and what kind of world is it operating in?



From PEAS to Rationality

In our previous lecture, we established how an agent interacts with the world.



Core Question

How does an agent choose actions that perform well?

This leads us to the concept of Rationality.

Learning Objectives

By the end of this lecture, you should be able to:

- Explain what a rational agent is.
- Understand the relationship between rationality and performance measures.
- Distinguish rational behavior from guaranteed success.
- Identify the main properties of an AI environment.

- Distinguish observable vs. partially observable.
- Distinguish deterministic vs. stochastic.
- Distinguish episodic vs. sequential.
- Distinguish static vs. dynamic.
- Distinguish discrete vs. continuous.
- Distinguish single-agent vs. multi-agent.

What Does "Rational" Mean?

Let's start with everyday intuition. You are driving to college.

> **Options:**
>
> **Route A:** 30 minutes
>
> **Route B:** 60 minutes

Assuming everything else is equal (safety, comfort), choosing **Route A** seems reasonable.

Why?

Because the action is expected to produce a **better outcome** according to your goal (arriving quickly). This is a **Rational Action**.



What Is a Rational Agent?

Formal Definition

A rational agent chooses the action that is expected to produce the best outcome according to its **performance measure** and **available information**.

How the decision is made:



Note: Rational agents do not need to know the future perfectly.

Rationality ≠ Perfection

This is a crucial misconception in AI. Let's look at what can happen:



Why can a rational agent fail?

- The environment is uncertain.
- Information may be incomplete.
- Sensors may be noisy or faulty.
- Future events cannot be known perfectly.

Key Takeaway

Rationality is about making the best decision with the information available ; not guaranteeing a perfect result.

Rationality Depends on the Performance Measure

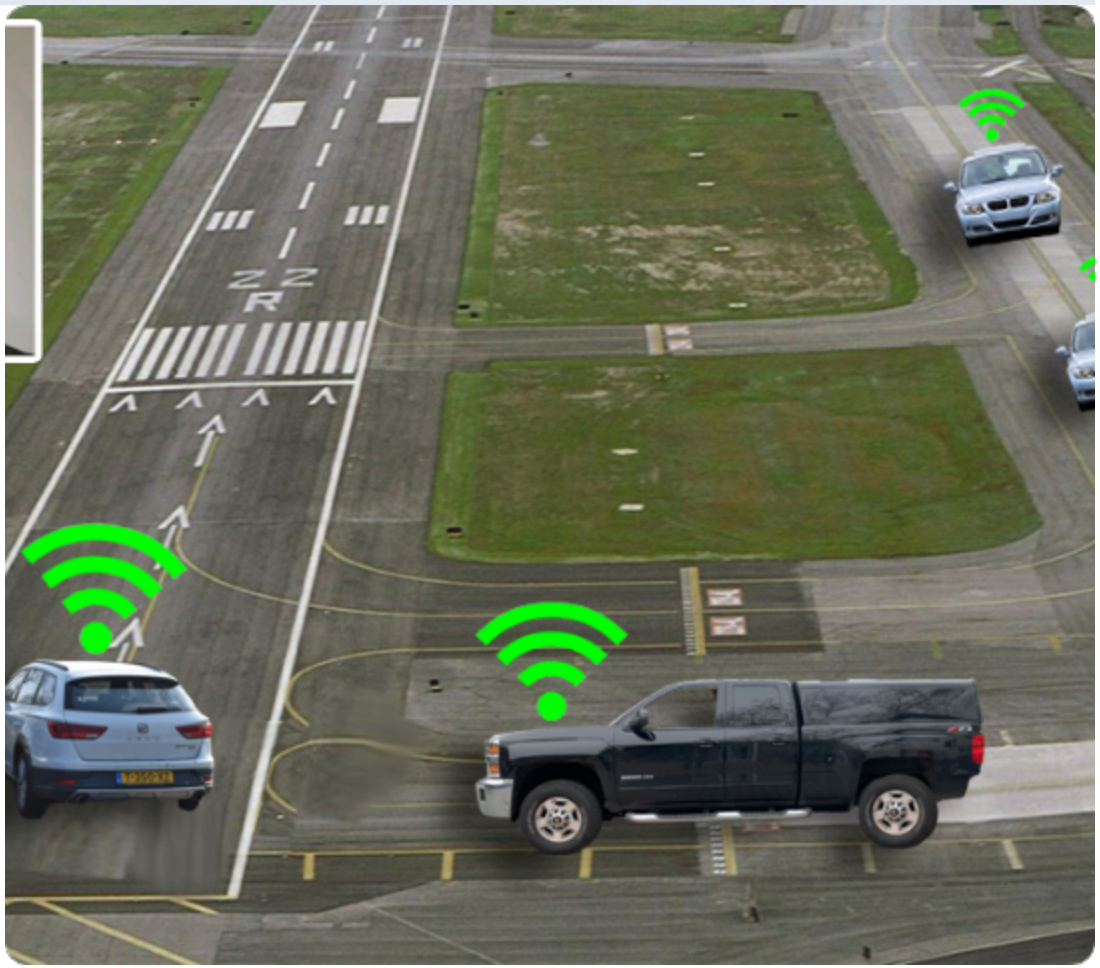
Consider a **Self-driving car** evaluating two actions:

Drive Faster vs Drive Slower

Which action is "rational"? It depends entirely on what we value:

- Travel time minimization?
- Absolute safety?
- Fuel/energy efficiency?
- Passenger comfort?

Rational behavior cannot be defined without knowing what counts as success (PEAS).



Rationality Depends on What the Agent Knows

Situation A

The Agent has:

- Clear, high-res sensors
- Complete information
- Reliable predictions of the future

Action chosen: High confidence, direct route.

Situation B

The Agent has:

- Noisy, obscured sensors
- Missing information
- Highly uncertain environment

Action chosen: Cautious, information-gathering route.

The rational decision can be different when the available information is different. Rationality is evaluated using **only the information available to the agent at that time**.

What Is an Environment?

> The **environment** is the world or situation in which an agent operates and with which it interacts.

Examples:

- **Robot vacuum:** The physical room.
- **Chess AI:** The chess board + opponent.
- **Self-driving car:** Road + traffic + weather.
- **Recommendation system:** Users + content + platform.

Emphasize: The environment does not have to be physical!



Why Do Environment Properties Matter?

Would the exact same AI programming strategy work equally well to play chess and to drive a self-driving car?

Obviously, no. Different environments have different characteristics that fundamentally change the problem. These characteristics affect:



What the agent can **observe**



How **predictable** the world is



Whether **other agents** are involved

Let's introduce the 6 primary environment properties.

1. Fully Observable vs Partially Observable

Fully Observable

The agent has access to **all relevant information** needed for the decision at any given time.



Example: Chess
The entire state of the board is visible to the agent.

Partially Observable

The agent **cannot access** all relevant information about the environment's state.



Example: Self-Driving Car
The agent cannot see around buildings or know other drivers' intentions.

Classifying Observability

Environment Task	Property	Why?
Chess	Fully observable	Board state is entirely visible at all times.
Tic-Tac-Toe	Fully observable	Entire 3x3 board is visible.
Driving a Car	**Partially observable**	Hidden obstacles, uncertain driver intentions, blind spots.
Poker	**Partially observable**	Opponents' cards and deck order are hidden.
Medical Diagnosis	**Partially observable**	Patient's true internal condition isn't directly visible.

**Note: Use "typically" or "generally", as classification can depend on the specific modeling assumptions of the system.*

2. Deterministic vs Stochastic

If you press the brake in a car, can you guarantee *exactly* what will happen to the millimeter?

🔗Deterministic

Given the current state and a specific action, the resulting next state is **entirely determined and predictable**.

Same situation + same action → **Same next state**

> **Example: Chess**
> Moving a piece follows strict rules. The outcome is 100% predictable.

🔀Stochastic

Outcomes involve **uncertainty or randomness**. The next state is not entirely predictable.

Same situation + same action → **Outcome may differ**

> **Example: Driving**
> Braking distance changes based on road dust, tire wear, or wind.

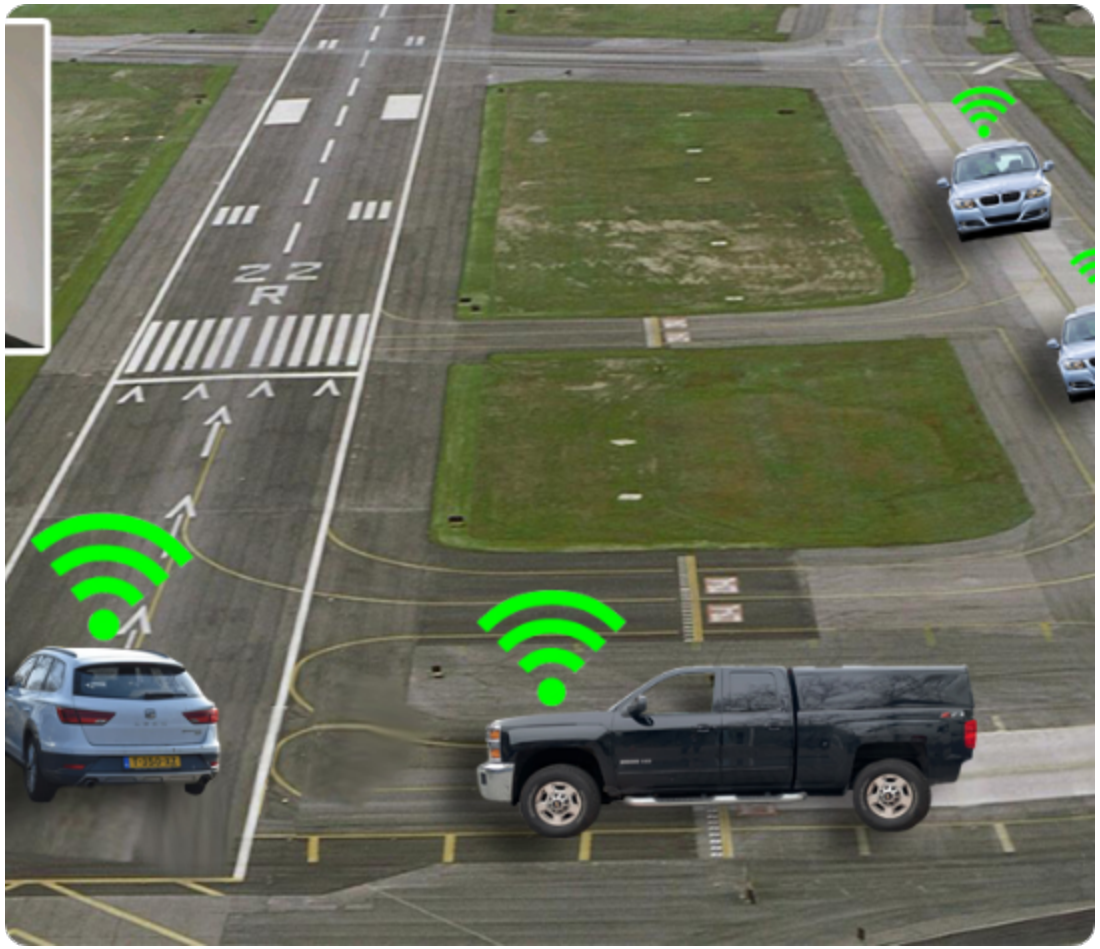
Why Does Uncertainty Matter?

Scenario: A self-driving car sees a pedestrian near a crosswalk. The car decides to gently brake.

The resulting outcome is **stochastic** because it depends on:

- Will the pedestrian suddenly run or stop?
- How will the vehicle behind react?
- Are the road conditions slightly slick?
- Sensor noise/uncertainty.

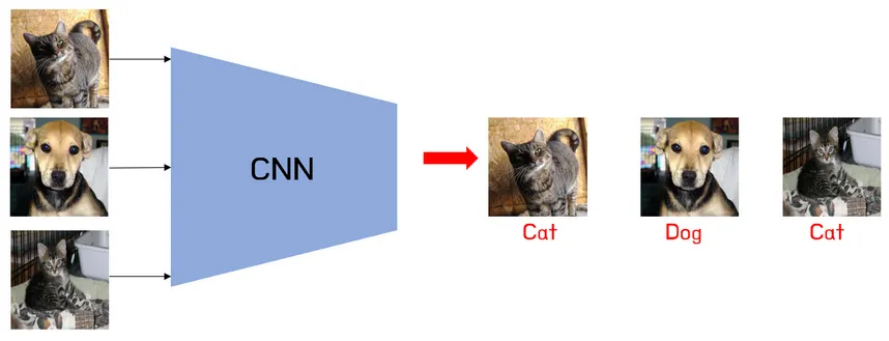
> **In stochastic environments, agents must make decisions under uncertainty, maximizing *expected* performance.**



3. Episodic vs Sequential

Episodic

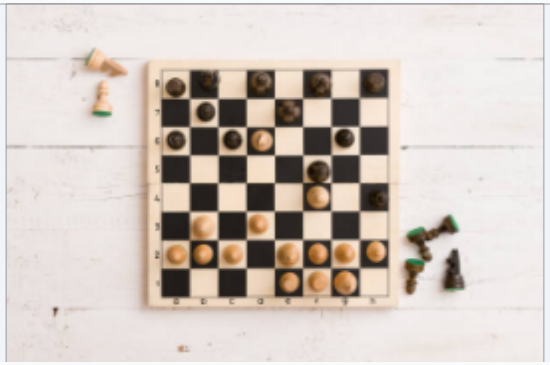
The agent's experience is divided into **independent episodes**. The current decision does not affect future episodes.



Example: Image Classification
Predicting "Dog" in one image doesn't change the next image in the folder.

Sequential

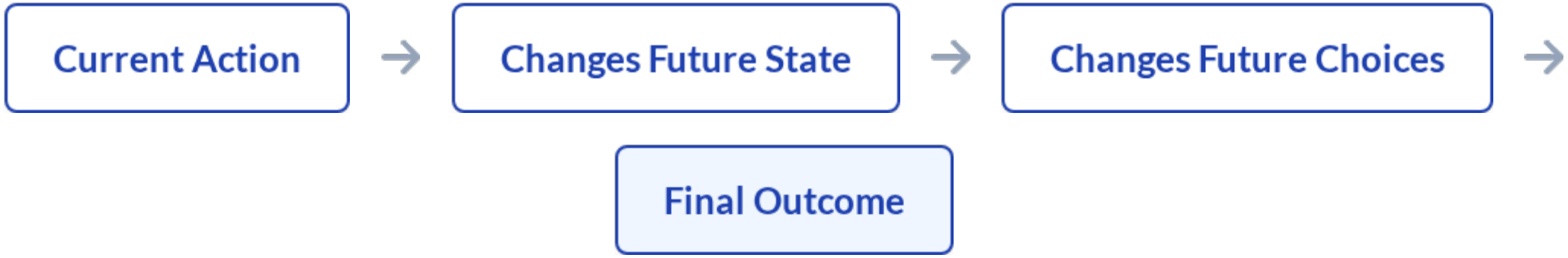
Current actions **affect future situations**. Decisions are linked in a chain.



Example: Chess
Sacrificing your queen now fundamentally changes the rest of the game.

Why Sequential Environments Are Harder

In a sequential environment, short-term actions have long-term consequences.



Chess Example:

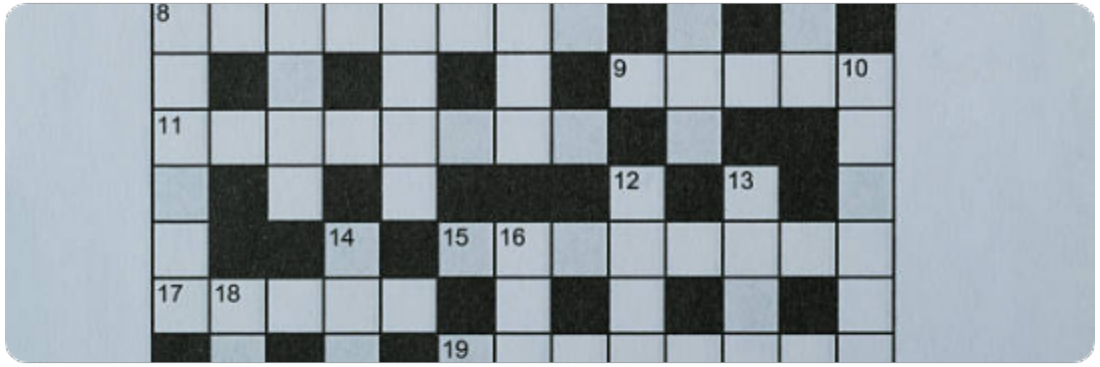
Move Knight → Opponent responds → New board state → Entirely new tree of possible moves → Eventual Win/Loss.

In sequential environments, an agent must "think ahead" and consider future consequences.

4. Static vs Dynamic

Static

The environment **does not change** while the agent is deliberating (thinking). The agent has all the time it needs.



Example: Crossword Puzzle

The words don't shift around while you think of an answer.

Dynamic

The environment **can change** while the agent is deliberating. Time matters.



Example: Driving in Traffic

If the AI takes 5 seconds to decide whether to brake, the situation has already changed dangerously.

Why Dynamic Environments Are Harder

In a dynamic environment, the "thinking" process itself consumes a valuable resource: **Time**.



Does a robot vacuum operate in a static or dynamic environment?

Typically Dynamic: Pets move, people walk, chairs get pushed while it cleans.

5. Discrete vs Continuous

Discrete

States, actions, and percepts can be represented as **distinct, countable choices**.



Example: Chess

There are a finite number of pieces.

The board has exactly 64 squares.

A move is discrete (e.g., e2 to e4).

Continuous

Variables take values over **continuous, effectively infinite ranges**.

Example: Driving

Speed: 45.234 mph

Steering angle: 12.5 degrees

Position and time flow continuously.

Why Does This Distinction Matter?

Discrete Environments

Often have countable states/actions.

Easier to represent explicitly in memory (like a vast tree or table).

Algorithms can "search" through explicit lists of possibilities.

Continuous Environments

Have huge or effectively infinite possibilities.

You cannot list every possible steering angle.

Require numerical measurements, approximations, and control theory.

Note: "Discrete" does not automatically mean "easy". Chess is discrete, but the number of possible games exceeds the number of atoms in the observable universe!

6. Single-Agent vs Multi-Agent

Single-Agent

Only **one agent** is the primary decision-maker in the environment. The environment itself is just the "world".

Example: Crossword Puzzle
You are solving it alone. The puzzle isn't fighting back.

Multi-Agent

Multiple agents interact in the environment simultaneously, reacting to each other.

Example: Chess (You vs. Opponent)
Example: Driving (Multiple vehicles)

Why is Multi-Agent harder?

Other agents can cooperate, compete, or behave unpredictably, requiring the AI to model *their* behavior.

Cooperative vs Competitive Agents

In multi-agent environments, relationships between agents dictate the strategy:

Cooperative

Agents share or support a common goal.



Example: Warehouse robots coordinating deliveries to avoid collisions and maximize throughput.

Competitive

Agents have conflicting goals (Zero-sum).



Example: Chess. Your goal (to win) directly conflicts with your opponent's goal (also to win).

Environment Properties Cheat Sheet

Property	Option 1	Option 2
Observability	**Fully:** See everything needed	**Partially:** Missing information
Predictability	**Deterministic:** Fixed outcome	**Stochastic:** Uncertain outcome
Dependency	**Episodic:** Independent actions	**Sequential:** Chain reaction
Change	**Static:** World waits for agent	**Dynamic:** World keeps moving
State/Action Space	**Discrete:** Countable choices	**Continuous:** Infinite ranges
Number of Agents	**Single-agent:** Acting alone	**Multi-agent:** Acting with others

Classroom Exercise: Classify These Environments

For each scenario, define the 6 properties: Observability, Determinism, Dependency, Change, State, Agents.

1. Chess

(Observable? Deterministic? Sequential? Static? Discrete? Single/Multi?)

2. Self-Driving Car

Classify all six properties. Which property makes this the hardest?

3. Robot Vacuum

Classify all six properties based on a typical household environment.

4. Medical Diagnosis System

Classify all six properties. Is the human body deterministic?

(Let's discuss these as a class before revealing the answers)

Final Recap + Exit Question

Rational Agent

Chooses actions expected to **maximize performance** given available information.

Rationality is **NOT** perfection or omniscience.

Environment Properties

- Observable?
- Deterministic?
- Episodic?
- Static?
- Discrete?
- Single-agent?

Exit Question

Why might a self-driving car require a *very different agent design* from a chess-playing AI?

Next lecture: Types of Intelligent Agents & Agent Architectures

Image Sources



https://media.easy-peasy.ai/f75451aa-55ed-4aee-8bc3-cc658a7000a1/839853e4-b779-4d2e-98df-74482f9a14f2_medium.webp

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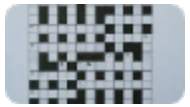
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